

Differentiation before frequency: Internal coordinative evaluation in linguistic stabilisation

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Abstract

Languages routinely generate structurally valid forms, yet grammatical derivation alone does not guarantee their stabilisation in usage. Existing linguistic accounts explain stabilisation through mechanisms such as frequency, entrenchment, propagation, and representational reinforcement, but leave comparatively underspecified how competing forms first become differentiated during language use. Drawing on a series of empirical case studies, primarily from Telugu, I argue that competing forms differ in how readily they integrate into ongoing production. On this basis, I propose internal coordinative evaluation, a production-level evaluative process through which competing forms acquire asymmetries in reproducibility prior to frequency-based amplification and entrenchment. This study reframes linguistic variation, stabilisation, and change as emerging in part from production-level differentiation and complements usage-based, exemplar-based, and competition-based approaches.

Keywords: Internal coordinative evaluation, Evaluative stabilisation, Linguistic stabilisation, Entrenchment, Usage-based linguistics, Phonological organisation.

1. Introduction

Languages are often taught as systems of rules. If a form follows the rules, it is considered correct, and if it violates them, it is incorrect. Yet language is not only a system of rules. It is also a system through which speakers continuously produce, perceive, and interpret linguistic forms in real time. Although grammatical rules determine what forms are possible, they do not determine which of these possibilities become established in everyday use. Not all correct forms survive, not all surviving forms are equally acceptable across contexts, and not all acceptable forms become equally stabilised across linguistic communities. Some forms feel natural and effortless, while others, though perfectly derivable, sound awkward, heavy, or strangely artificial. Speakers may understand them, but hesitate to use them. Over time, such forms may retreat into restricted registers or disappear altogether. Grammatical correctness alone therefore does not explain why some constructible forms become stable while others remain marginal.

This raises a fundamental question: what determines which constructible alternatives become reproducibly stabilised within a language? Consider plural formation in Telugu. Grammar provides a clear mechanism in which a noun combines with a plural marker, and the resulting form should in principle be acceptable. On paper, the process is straightforward and productive. Yet in practice speakers consistently prefer some plural forms while avoiding others that are equally constructible from the same morphophonological resources. The rejected forms are not unintelligible nor ungrammatical. They simply fail to be consistently reproduced during ordinary language use. Comparable tensions between constructibility, variation, coexistence, and absence have also been documented in research on morphological competition, overabundance, and defectivity, where grammatically available forms do not necessarily stabilise uniformly in usage (Thornton, 2011; Bermel and Nikolaev, 2022; O'Neill, 2025). This recurrence across typologically distinct languages suggests that the responsible forces operate not within any particular grammar but within the organisation of language use itself, extending the problem beyond Telugu to a more general question concerning linguistic stabilisation.

This contrast between what can be formed and what ultimately stabilises has long been difficult to explain within linguistic theory. Traditional grammatical approaches distinguished between competence and performance, effectively placing questions of usage outside the explanatory scope of grammar itself (Chomsky, 1965). Subsequent work acknowledged that grammatical structure alone cannot account for patterns of frequency, entrenchment, or long-term survival (Newmeyer, 1983). These developments opened the door for cognitive and usage-based models of language. More recent developments have further expanded these perspectives through accounts of propagation, conventionalization, constructionalisation, diffusion, and morphological competition, emphasizing how linguistic forms circulate, stabilise, and coexist across speakers and systems (Milroy, 1992; Traugott and Trousdale 2013; Schmid, 2020; Bermel and Nikolaev, 2022).

Within cognitive linguistics, linguistic structure is understood as emerging from patterns of usage and experience. Many usage-based models emphasise mechanisms such as entrenchment, exemplar accumulation, schema formation, and network-based emergence (Langacker, 1987; Bybee, 2001, 2010; Pierrehumbert, 2001; Schmid, 2007, 2020; Wedel, 2006; Diessel, 2019). Repeated exposure increases the frequency of particular forms, frequency in turn strengthens their cognitive representation through entrenchment, and these strengthened representations increase the likelihood that those forms will be reproduced in subsequent use. These approaches thus provide compelling explanations for how linguistic asymmetries become reinforced once they have emerged. They are less explicit, however, about how competing grammatical alternatives first acquire those asymmetries before substantial differences in frequency or entrenchment exist. Frequency here functions both as the explanation these models provide and as the phenomenon that itself requires explanation. If entrenchment is driven by repetition, and repetition is driven by frequency, the account then risks circularity unless the origin of frequency differences is independently addressed.

Research in phonetics and speech perception provides a part of the answer to this question. Linguistic forms are not only represented as abstract structures but are experienced

as acoustic and articulatory events (Browman and Goldstein, 1992; Fowler, 1986). Speech production and perception involve the real-time coordination of motor gestures, auditory expectations, and phonological patterns. As a result, some sequences integrate smoothly into ongoing speech, while others introduce greater coordination demands, even though both satisfy grammatical constraints. Such differences indicate that grammatically possible alternatives may vary in how readily they function within ongoing speech processes, even though speakers themselves are typically unaware of these dynamics during production. What is striking here is that speakers do not consciously calculate syllable counts or articulatory effort when producing a form. Rather, these pressures operate automatically during speech processing and may contribute to differences in reduction and recurrence over time (Lindblom, 1990; Ohala, 1993; Kirchner, 1998). Such work helps explain why some forms become easier to reproduce, but it does not fully explain how competing grammatical alternatives first become differentially reproducible before later frequency-based reinforcement.

A similar emphasis on propagation appears in work on language change. Classical formulations addressed the “evaluation problem”, asking how linguistic forms acquire value within speech communities (Weinreich et al. 1968). Later evolutionary models described the spread of forms as a process of selection operating across populations of speakers (Croft, 2000), often emerging from the cumulative consequences of individual linguistic behaviour in what has been termed an “invisible hand” process (Keller, 1994). Once forms begin to propagate, their trajectories frequently follow characteristic diffusion patterns such as the S-curve widely observed in language changes (Blythe and Croft, 2012). These approaches explain how forms spread after they have entered circulation, but they leave open the earlier point at which competing forms first begin to diverge during language use itself.

The present study targets precisely this earlier stage. It begins not with a theory, but with a set of recurring tensions that grammar leaves unresolved across morphology, register variation, lexical competition, dialect contact, borrowing, and everyday speech production. Across these domains, competing forms repeatedly exhibit unequal reproducibility despite remaining grammatically possible. On this basis, I propose Internal Coordinative Evaluation (ICE), a production-level process through which co-possible forms acquire asymmetries in reproducibility before frequency-based amplification and entrenchment. This perspective seeks to explain why some words feel complete while others remain persistently awkward, why identical morphophonological resources behave differently across lexical and social contexts, and why speakers often converge on similar forms without explicit instruction or awareness. The following sections first establish this empirical pattern and then develop the theoretical account proposed to explain it.

2. Methods

2.1 Participants

A total of 154 Telugu speakers, ranging in age from 5 — 70 years, participated across the different components of this study. The primary dataset consisted 114 speakers. Two additional

regional comparison datasets comprising 19 and 21 speakers, respectively, were recruited for the dialect analyses. The sample included both literate and non-literate speakers, as well as a small number of bilingual individuals. Participants were drawn from everyday social environments, including households, schools, and public places, and were selected opportunistically to reflect a range of age groups and backgrounds. All participants provided informed consent. The study involves minimal-risk linguistic data and did not require formal institutional ethics approval.

2.2. Elicitation tasks

Data were collected through brief elicitation tasks conducted in both formal (schools) and informal settings. Participants were asked simple production questions such as ‘What is the plural of X?’ or ‘Which form would you use?’ for a set of familiar lexical items. Instructions emphasized spontaneous responses, with participants encouraged to produce forms as they would in everyday speech without deliberation. For responses that showed strong convergence, follow-up questions were asked to probe acceptability of alternative forms. These included prompts such as whether the alternative form sounded acceptable, incorrect, or unusual. Follow-up questions were only asked after the initial response had been provided and did not involve prompting toward any particular answer. In cases involving regional or contact forms, participants’ familiarity with alternative forms and their judgements about naturalness were also recorded.

2.3. Procedure

Each session was conducted individually or in small group settings and typically lasted 2 — 3 minutes. Participants were informed that they could decline to respond to any question. Responses were collected in naturalistic environments, including homes, schools, and public locations. No time constraints were imposed beyond encouraging spontaneous responses. However, younger participants occasionally exhibited brief response delays when producing forms that were not part of their entrenched usage.

2.4. Data recording

Responses were recorded manually during elicitation. For each item, the form produced by the participant was noted. In addition, instances of correction (e.g., adult correction of children’s responses) and qualitative judgments about alternative forms (e.g., perceived unnaturalness or acceptability) were documented. In a subset of cases involving younger participants, brief response delays were noted qualitatively. The data are exploratory, and therefore no precise timing measures were used.

2.5. Analysis

The analysis focuses on distributional patterns across participants. For each item, the proportion of speakers producing a given form was calculated. In cases of near-uniform convergence, patterns are described categorically. The goal of the analysis is not statistical inference but to identify robust patterns of convergence and divergence in production across speakers with varied backgrounds and exposure histories.

3. Case Studies

3.1. The same suffix, different fates

One of the most revealing places to look for structure in a language is not in rare or exotic constructions, but in the most ordinary ones. In Telugu, plural formation offers exactly such a window. At first glance, the plural marker -lu (/lu/) appears straightforward. Yet, when the same suffix attaches to different singular forms, it often produces sharply different outcomes. Some plural forms stabilise effortlessly and feel native. Others remain marginal, restricted to specific registers. Still others, though constructable from the same morphophonological resources, fail to gain traction altogether.

Let us consider a basic singular noun such as *ceṭṭu* /tʃɛʈʈu/ ('tree'). When the plural marker -lu is applied, it does not yield a single stable form, but several competing ones such as *ceṭlu* /tʃɛʈlu/, *ceṭṭulu* /tʃɛʈʈulu/, and *ceṭṭul* /tʃɛʈʈul/, all of which are derivationally plausible within the morphophonological resources of the language. However, speaker elicitation and corpus data (Table 1a) show that all speakers converge on *ceṭlu* as their spontaneous production reference. A small percentage of speakers (~10%) acknowledged *ceṭṭulu* as an acceptable alternative but still produced *ceṭlu* as their first response. This distinction between produced form and accepted alternative is important. A form can be judged acceptable without being produced as the default, which suggests that something beyond grammatical judgment is conditioning production preferences. The suffix remains constant across these outcomes. What changes is how it combines with the base.

This pattern shows that plural formation does not yield equivalent outcomes across co-possible forms. Research on usage-based morphology has shown that frequency, repetition, and articulatory ease influence which forms become entrenched (Bybee, 1985; Hay, 2001). However, these correlations do not explain why one form becomes dominant among co-possible alternatives at the point of first use. The cases below examine this selective behaviour directly.

3.1.1. *paṇḍu* to *paṇḍlu* / *paṇḍlu*, but not *paṇḍulu*

From a grammatical or morphophonological point of view, nothing in the plural template appears to prevent the formation *paṇḍulu* /paṇḍulu/. The singular form *paṇḍu* /paṇḍu/ ('fruit') ends in a vowel, and the plural suffix -lu should attach without difficulty. Yet, *paṇḍulu* remains absent in ordinary conversational usage. Combined elicitation and corpus data from adult speakers (Table 1a) show that a majority of speakers (~74%) produce *paṇḍlu* as their preferred plural form, with the remainder (14%) converged on *paṇḍlu* /paṇḍlu/. A small proportion of speakers accepted *paṇḍulu* as recognizable forms, but none produced them in everyday speech.

The direction of this pattern is significant. *Paṇḍlu* reduces syllabic weight and produces a tighter consonantal alignment than *paṇḍulu*. *Paṇḍlu* compresses further through lateral assimilation, producing a prosodically unified output. Both preferred forms achieve a more compact structure than that yielded by transparent suffixation. The rejected form *paṇḍulu* is

not unclear or ill-formed. It remains interpretable but does not circulate in use. This contrast suggests that plural formation does not yield uniform outcomes across co-possible forms.

Table 1a. Plural formation: *paṇḍu* (adult speakers; n = 114; alternatives: *paṇḍulu*, *ceṭṭulu*)

Singular form	Preferred plural form	Speaker convergence (%)	Alternative accepted but not produced
<i>paṇḍu</i>	<i>paṇḍlu</i>	73.70	7 speakers (8.30%) accept <i>paṇḍulu</i>
<i>paṇḍu</i>	<i>paḷḷu</i>	14	3 speakers (18.80%) accept <i>paṇḍulu</i>
<i>ceṭṭu</i>	<i>ceṭḷu</i>	100	12 speakers (10.50%) accept <i>ceṭṭulu</i>

Note: Percentages in the final column are calculated relative to the subgroup of speakers who produced the corresponding preferred plural form in that row (e.g., $7/84 \approx 8.30\%$ for the *paṇḍlu* row), not to the full sample of n = 114.

Table 1b. Plural formation: *paṇḍu* (younger speakers; n = 31)

Singular form	Produced plural form	Speakers (%)
<i>paṇḍu</i>	<i>paṇḍulu</i>	12.90
<i>ceṭṭu</i>	<i>ceṭṭulu</i>	12.90

Additional elicitation from young speakers further illustrates the pattern observed across the dataset. As shown in Table 1b, a subset of younger speakers (~13%) produced *paṇḍulu* and *ceṭṭulu* after a brief hesitation. The response was immediately corrected by adult caregivers, who consistently preferred *paṇḍlu*, and *ceṭḷu*, but rejected *paṇḍulu* and *ceṭṭulu* as incorrect. The lack of persistence of these forms in adult usage indicates that not all constructible outputs are sustained in the speech community. This contrast between initial production and subsequent correction highlights a gap between what can be generated and what is maintained in use.

3.1.2. *pannu* splits: *paḷḷu* and *pannulu*

An even more revealing case of the selective behaviour involves the surface form *pannu* /pannu/. The word can mean ‘tooth’ or ‘tax’, depending on context, yet their plural outcomes diverge categorically. For ‘tooth’, speakers produce *paḷḷu* /paḷḷu/, not *pannulu* /pannulu/. For ‘tax’, the preferred plural is *pannulu*, not *paḷḷu*. Combined elicitation and corpus data show that all speakers (100%) converged on *paḷḷu* for the ‘tooth’ meaning, and produced *pannulu* for the ‘tax’ meaning.

This contrast cannot be explained by phonology alone, since the singular form is identical. Nor can it be explained by morphological rule since the same resources apply to both meanings. What differs is the domain of use in which each meaning circulates. Corpus data show that the ‘tooth’ sense occurs overwhelmingly in conversational registers, while the ‘tax’ sense belongs to an abstract, formal domain where the expanded form fits the deliberate pace of institutional contexts. Studies of lexical diffusion and paradigm uniformity have shown that frequency and domain of use can condition morphological outcomes (Phillips 1984; Bybee 2007). However, these accounts do not explain why identical forms yield different outcomes across contexts. The present case highlights this contrast.

3.1.3. Why *ēḍu* avoids *ēḍulu*

A final case sharpens the point further. The word *ēḍu* /e:ḍu/ (‘year’) does not pluralise as *ēḍulu* /e:ḍulu/, even though this would be the transparent suffixation outcome. Nor does it become *ēḍlu* /e:ḍlu/, by analogy with forms such as *paṇḍlu*. However, speaker data are unambiguous. No speaker in the elicitation study preferred *ēḍulu* as the plural of *ēḍu*, and *ēḍulu* is absent from the naturalistic corpus data. The categorical rejection of constructible output, not merely its dispreference but its complete non-occurrence, provides the clearest evidence that constructability does not guarantee distributional access.

Instead, speakers produce *ēṇḍlu* /e:nḍlu/ or *ēḷḷu* /e:ḷḷu/. Both involve internal restructuring rather than simple suffixation. The insertion of a nasal in *ēṇḍlu* creates a more compact coda cluster. Lateral assimilation in *ēḷḷu* produces a prosodically unified output. In both cases the cognitive system reorganises the word to achieve a form that integrates more readily in production rather than following the transparent derivational path. Crucially, *ēḍulu* has zero frequency not because speakers encountered an alternative form more often and entrenched it, but because *ēḍulu* never stabilised in the first place. The asymmetry precedes any frequency difference rather than resulting from it.

Across these cases, a convergent pattern emerges. Forms produced by transparent suffixation do not stabilise, even when they are constructable. Alternative forms involving compression or restructuring circulate more widely. The same suffix yields different outcomes depending on the lexical item and its usage context. Speakers converge on these outcomes with remarkable consistency across age, gender, and literacy level, including among non-literate adults and young speakers who have had no exposure to formal grammar instruction. This cross-demographic uniformity suggests that the preferences documented here are not the product of schooling, prescriptive norms, or social identity. Thus, plural formation, often treated as routine, turns out to be a sensitive diagnostic for observing how co-possible forms diverge in use. However, the pattern observed here prompts a further question. If constructability is not sufficient for distributional success, what determines which forms stabilise?

3.2. Selective survival across registers within a single language

The previous case showed that grammatical derivation alone does not determine which forms survive in a language. The same pattern becomes even clearer when we look not across different languages, but within the same language, across different registers such as everyday speech, poetry, and formal usage.

In everyday spoken Telugu, language operates under constant demands of speed, interaction, and real-time coordination. Words must fit smoothly into ongoing speech without disrupting timing or articulatory coordination. Forms that introduce extra syllabic weight remain limited in spontaneous speech, not because they are unintelligible, but because they introduce additional structural load during production. Speaker elicitation data in Table 1a show that all speakers (100%) converge on *ceṭṭu* as their spontaneous plural of *ceṭṭu* in conversational contexts, with only a small minority (~10%) acknowledging *ceṭṭulu* as a possible alternative, and none producing *ceṭṭulu* as their first response in the conversational condition. The distinction between passive acceptability and active production preference is directly relevant here. A form can be judged possible without being produced as the default. This contrast shows that constructibility alone does not determine production in spontaneous speech. Similar observations have been made in research on speech production where sequences that integrate smoothly into ongoing articulation tend to recur more readily than those that interrupt planning (Levelt, 1989; Lindblom, 1990).

Poetic language does not eliminate constraints. Instead, it reorganises them. While everyday speech favours compactness and processing efficiency, poetry operates under metrical and rhythmic demands. Form such as *ceṭṭul* and *ceṭṭulu* are both attested in several poetic usages, but their distribution depends on metrical structure. A form such as *ceṭṭul* fits a two-beat structure while retaining a consonantal core, allowing it to anchor a metrical position within a line. By contrast, *ceṭṭulu* introduces additional syllabic weight, making it more suitable in contexts where a longer rhythmic unit is required. The same forms that are limited or dispreferred in everyday speech may therefore align more effectively within poetic rhythm. Their occurrence in verse therefore reflects context-dependent suitability within metrical structure rather than unrestricted variation in a different processing environment (for general discussion of metrical constraints, see Fabb, 1997). Systematic corpus evidence for the distribution of these forms across metrical contexts remains a direction for future research.

Formal usage introduces yet another set of conditions. In this domain forms persist not because they integrate naturally into speech or rhythm, but because they are maintained through education, standardisation, and textual transmission. A within-speaker elicitation design asked the same speakers which form they would use in conversation and then which form they would use in formal or written contexts. The results show a clear shift across contexts. Speakers consistently produced *schoolḷu* /sku:ḷlu/ ('schools') in the conversational condition, while almost all speakers (98%) produced *pāṭhaśāḷalu* /pa:ṭʰaʃa:lalu/ in formal contexts. For vehicles, *baṇḍlu* /baṇḍlu/ was the conversational form, while *vāhanālu* /va:ʱana:lu/ was the preferred formal alternative. No speaker produced *vāhanālu* spontaneously in the conversational condition. The within-speaker nature of this contrast is theoretically important. The difference

does not reflect two separate populations with different linguistic preferences. The same individuals prefer different forms depending on the conditions active at the moment of production rather than relying on a single fixed lexical choice.

Formal plural forms such as *pāṭhaśālalu* and *vāhanālu* do not circulate widely in everyday speech even though they remain part of the recognized vocabulary. Their continued presence appears to depend on institutional support through education, literacy, and standardisation. This allows forms that are less compatible with spontaneous speech to remain available within formal registers. Similar patterns of register stratification have been documented in sociolinguistic research on standardisation (Milroy and Milroy, 1999).

Across these three registers, the morphophonological system remains constant. What changes are the conditions under which forms are processed. A form that circulates widely in conversation may be structurally unsuited to poetry, and a form preserved in formal writing may remain peripheral in everyday speech. Registers therefore provide distinct contexts in which the same constructible forms show different distributional outcomes. The outcomes remain consistent within each domain even though the underlying morphophonological resources remain unchanged. If the same resources yield different outcomes depending on context, then the source of variation cannot be explained by the morphophonological resources alone. This raises two questions. What determines which forms stabilise across these contexts, and does this pattern extend beyond morphology and register within a single language to cases where distinct phonological systems come into contact?

3.3. Across borders: Dialect, contact, and structural repair

The patterns observed so far show that constructible forms do not circulate equally within a single language. When we turn to dialect contact, a related but distinct pattern emerges. Instead of merely restricting forms to particular domains, contact settings often reshape them. This reshaping is not random, but tends to follow structural patterns while preserving the original meaning.

In some Telugu-speaking regions, particularly those closer to Tamil-speaking areas, speakers commonly produce *kōsaram* /ko:saram/ where Standard Telugu uses *kōsam* /ko:sam/, meaning ‘for’ or ‘for the sake of’. Elicitation data from nineteen regional speakers in Tamil-Telugu contact zones show that these speakers find *kōsaram* natural for everyday speech in their region, while also recognising *kōsam* as an acceptable alternative. When *kōsaram* was presented to speakers from other Telugu-speaking communities, none selected it as their preferred form and several described it as unusual or incorrect. This between-community contrast is central. The same meaning is expressed through different forms in different communities, and each community evaluates the other form as marked.

Table 2: Contact variation: *kōsam* vs *kōsaram* (contact zone speakers; n = 19)

Speaker group	Preferred everyday form	Speakers (%)	Alternative form recognised	Evaluation of alternative form
Tamil-Telugu contact zone	<i>kōsaram</i>	100	<i>kōsam</i>	Acceptable but not preferred
Non-contact Telugu communities	<i>kōsam</i>	100	<i>kōsaram</i>	judged as inappropriate

Table 2 shows that speakers differ not only in which forms they produce in everyday speech, but also in how they evaluate alternatives when these are introduced. Notably, non-contact speakers rejected *kōsaram* despite understanding its meaning. This contrast shows that interpretability does not guarantee acceptance or production.

One possible explanation for this divergence lies in the phonological coordination patterns across speech communities. The form *kōsam* ends in a fricative plus nasal sequence that integrates smoothly in Standard Telugu. In contact regions influenced by Tamil, where disyllabic word-final shapes such as *-aram* and *-uram* are frequent, *kōsaram* aligns with patterns that are more common in those environments. Similar forms of adaptation have been documented in studies of language contact, where forms are reshaped as they circulate across linguistic systems (Thomason and Kaufman, 1988; Heine and Kuteva, 2005; Matras, 2009).

Crucially, the restructuring does not appear to result from conscious planning. Speakers do not deliberately insert segments when producing *kōsaram*, yet speakers within the contact zone converge on a similar outcome. If the change were purely accidental or imitative, wider variation would be expected. Instead, outcomes are relatively stable within the region. The meaning of *kōsam* is preserved entirely in *kōsaram*, while only the form differs. Comparable observations appear in studies of phonological adaptation, where borrowed forms retain semantic identity while undergoing structural change (Kang, 2011).

This is not an isolated phenomenon. The form *avasaram* /*avasaram*/ (‘necessity’), ultimately traceable to Sanskrit *avaśyam* /*avaɕjam*/, shows a comparable expansion that has become part of Standard Telugu. Unlike *kōsaram*, which remains regionally distributed, *avasaram* eventually stabilised across speech communities. Historical linguistics documents many cases in which forms shaped through contact or local variation later spread more widely once they become established in usage, often through processes of diffusion and entrenchment (Hock, 1991; Bybee, 2015). The contrast between *kōsaram* and *avasaram* suggests that similar contact histories need not produce similar distributional outcomes.

3.4. Lexical competition beyond affixation

The patterned differentiation observed in morphological reshaping is not limited to affixation. Similar patterns also appear in cases of lexical rivalry, where semantically equivalent forms exhibit divergent distributional trajectories. Consider the lexical equivalents for ‘pillow’ in

Telugu: *talagaḍa* /talagaḍa/, *metta* /met:a/, and *dinḍu* /dinḍu/. All three forms refer to the same object and are recognised within the language. However, their distribution is not uniform.

Elicitation data across Telugu-speaking communities show that *dinḍu* is widely accepted across regional groups (Table 3). By contrast, *metta* and *talagaḍa* are each preferred mainly within particular regional communities and are evaluated negatively outside them. While some speakers who prefer *talagaḍa* described *metta* as unnatural or unfamiliar, 36 speakers from Community B who prefer *metta* described *talagaḍa* as sounding unfamiliar. Nevertheless, *dinḍu* was accepted by all groups, including those who primarily use *metta* or *talagaḍa*.

Table 3: Lexical rivalry: pillow alternative forms across Telugu regional communities

Regional Telugu-speaking community	Primary lexical form	Evaluation of competing regional form	Response to <i>dinḍu</i>
Community A (n = 21)	<i>talagaḍa</i>	<i>metta</i> described as unknown	Accepted as natural
Community B (n = 114)	<i>metta</i>	<i>talagaḍa</i> described as unnatural or unfamiliar (n = 36)	Accepted as natural

Structurally, the three forms differ in syllabic span and internal configuration. While *talagaḍa* extends across four light syllables, *metta* contains a medial geminate that creates an abrupt interruption in the syllabic flow. But *dinḍu* forms a compact disyllable with internal nasal-stop closure that resolves cleanly. The cross-regional acceptance of *dinḍu* shows that some forms circulate more widely across communities than others. By contrast, *metta* and *talagaḍa* are largely confined their respective regional systems. Speakers often respond to unfamiliar forms by describing them as unusual or incorrect. This contrast is observed even in the absence of semantic or grammatical differences.

Multiple constructible forms coexist, but their distribution is uneven. Semantically equivalent and grammatically possible alternatives may therefore exhibit markedly different patterns of usage, with some becoming widespread while others remain regionally restricted.

3.5. Distant languages, similar repairs

The preceding examples have all been drawn from Telugu and its immediate linguistic environment, spanning morphology, register variation, dialect contact, and lexical competition. Although these observations reveal a consistent empirical pattern, they could still be interpreted as language-specific phenomena. The question therefore arises whether comparable patterns

also appear in historically and geographically unrelated languages. Loanword adaptation provides an opportunity to examine this possibility.

When English words enter Japanese, they undergo systematic reshaping. Consonant clusters are broken apart, vowels are inserted, and word-final consonants are avoided. Japanese syllable structure is predominantly consonant-vowel-based and organized around mora timing, which places strong restrictions on complex codas and consonant clusters. As a consequence, the English word ‘strike’ becomes *sutoraiku* /sutoraiku/ in Japanese. Similarly, ‘club’ becomes *kurabu* /kurabu/ and ‘glass’ becomes *gurasu* /gurasu/. These transformations are often explained as a consequence of phonotactic restriction. Because Japanese does not permit certain consonant clusters, vowels are inserted to produce pronounceable sequences. While this explanation captures an important part of the process, it leaves another question open. Phonotactic restrictions explain why ‘strike’ cannot remain unchanged, but they do not fully determine which adapted form will be used. In practice, the borrowed form does not circulate in multiple unstable forms. Instead, speakers converge on a particular reshaped form.

This reshaping is neither arbitrary nor unlimited. The resulting form consistently conforms to Japanese phonological organisation. Consonants are separated by vowels, syllables align with preferred timing units, and the word is given a stable ending that integrates smoothly into Japanese speech. Studies of loanword phonology have shown that such adaptations follow structural patterns of the recipient language rather than preserving the phonetic surface of the source (Kang, 2011). This pattern resembles the earlier example in which *kōsam* becoming *kōsaram*. In both cases the incoming form is not rejected but reorganised in ways that match the structural patterns of the receiving system.

Comparable patterns have been documented across languages that restrict complex syllable structures, where loanwords are systematically reshaped through processes such as vowel insertion and syllabic expansion. In such cases, incoming forms are reorganised to conform to the phonological organisation of the recipient system rather than preserving the phonetic surface of the source. Cross-linguistic studies of loanword adaptation consistently show that such restructuring follows the phonological organisation of the system, including its constraints on syllable structure and segment sequencing (Itô and Mester, 1995).

These observations suggest that reshaping is not confined to a single language or region. Comparable responses emerge under different conditions, including register variation, dialect contact, and foreign input, although the specific outcomes differ across linguistic systems. The Japanese examples therefore extend the empirical patterns beyond Telugu, indicating that comparable forms of restructuring can arise in historically and geographically unrelated languages.

3.6. Asymmetries in the unfolding of speech

The patterns documented so far concern how forms survive, spread, or disappear across time and across communities. The same pattern can also be observed at a much smaller temporal scale: the moment of speaking itself. Long before a form becomes dominant across regions or

stabilises in literature, differences are already observable in real time. Some sequences pass through speech without disruption, while others require adjustment, and this asymmetry becomes apparent in the flow of speech itself.

Consider the contrast between Telugu phrases *evari kōsam* and *evari kōsaram*, both meaning ‘for whom’. Both are meaningful and are used by speakers in different regions. Yet in rapid everyday speech, *evari kōsam* fits more compactly within the surrounding sequence. The longer form adds an additional syllable without altering the meaning. This difference becomes more noticeable when the phrase is repeated or extended within rhythm or melody. Substituting *kōsaram* in such contexts increases syllabic weight and can disrupt the rhythmic contour of the phrase under rapid articulation. The difference is therefore not ideological, yet it is observable within the timing of speech itself. Speakers do not arrive at this preference by discussion or explicit agreement. Even speakers who identify with the longer form may shift toward the shorter form under conditions of speed or rhythm. Similar patterns have been noted in research on speech production, where differences in sequencing and timing can affect how utterances unfold (Levelt, 1989). This differs from the between-community patterns discussed earlier, where different groups favour different forms. Here, variation can occur within the same speaker depending on the conditions of use.

A similar phenomenon appears in Telugu expressions such as *undi kadā* and *vundi gada*, both conveying ‘it is there, isn’t it?’. Both are widely understood, yet some speakers report that *vundi gada* transitions more readily in rapid speech. The consonantal onset *vu* /v/ allows a smoother transition into the phrase than the vowel-initial *undi kadā*, and the transition from *di* to *ga* may involve less articulatory adjustment than the transition from *di* to *ka* in that environment. This does not imply that *ga* is generally more preferred to *ka*. In other environments, such as *ennō kānukalu* (‘many gifts’), replacing the *ka* with *ga* would immediately disrupt the sequence. The contrast therefore depends on the local configuration of the phrase rather than on a general preference for particular segments. Related observations appear in studies of coarticulation, where local phonetic configurations influence how sequences are realised in connected speech (Fowler, 1986). A comparable contrast appears in Telugu expressions such as *ēmi cēstunnāvu* and *ēm jēstunnāvu* (‘what are you doing?’). Even when the latter is treated as regional or informal, the shift from *cē* to *jē* produces a more compact sequence in rapid speech. Although such forms may later attract social interpretations, the difference first becomes visible in the timing of production.

Across these cases, a consistent pattern appears. Some forms pass through repeated use without disruption, while others require adjustment to remain stable. Speakers do not appear to compare alternatives explicitly when producing these sequences. Although the competing forms remain equally available within the grammatical resources of the language, they do not behave equivalently during production. The same pattern observed across the preceding domains is also evident in within the unfolding of speech itself. The recurrence of this pattern across multiple linguistic domains suggests that these observations are unlikely to represent isolated phenomena, but instead reflects a broader empirical regularity that calls for explanation.

4. From pattern to process

Across the domains examined here, a recurrent empirical pattern becomes difficult to dismiss. Forms do not simply appear and disappear at random, nor do they persist solely because speakers endorse them or institutions promote them. Multiple grammatically possible alternatives coexist within the resources of a language, yet their distribution is rarely neutral. Certain configurations circulate more readily, others remain restricted, and still others undergo structural reshaping before stabilising. These differences become visible in the dynamics of actual speech, where some forms integrate more readily into ongoing production while others require additional adjustment to remain stable.

Taken together, these observations suggest that differentiation arises neither solely from individual preference, regional identity, or institutional alignment, nor solely from the cumulative strengthening of representations through frequency. Forms stabilise because, in use, some configurations consistently function more readily under the relevant structural conditions than competing alternatives.

Existing linguistic approaches provide strong accounts of how patterns spread and consolidate once differences in usage have emerged. The cases examined here instead point to an earlier stage at which differentiation among co-possible forms becomes detectable before substantial frequency divergence or institutional standardisation. If stabilisation is structural rather than semantic or social, if grammatical generation alone does not determine circulation, if pressure-based accounts primarily explain tendencies in realisation, and if frequency reflects accumulated reproduction rather than initial differentiation, then the point at which such alternatives first become asymmetrical must lie elsewhere in the process of language use itself.

On this basis, I propose that a process operates during language use through which competing forms become differentiated prior to substantial frequency divergence and entrenchment. I refer to this process as Internal Coordinative Evaluation (ICE). The following section develops its ontological status and mechanism in full. Here, it is sufficient to propose that ICE introduces asymmetries in reproducibility during enactment, prior to substantial frequency divergence.

5. Internal Coordinative Evaluation: A theoretical account

5.1. *Ontological Status*

Internal Coordinative Evaluation (ICE) refers to an automatic process of coordinative evaluation arising during production, whereby competing phonological candidates are assessed relative to the structural organisation of the linguistic system, resulting in subtle asymmetries in their coordinative compatibility within ongoing language use. The term ‘coordinative’ is used here in a system-level sense, referring to the integration of multiple processing pressures in the course of production rather than to gestural coordination in articulatory phonology (Browman and Goldstein, 1992).

ICE is an emergent property of coordinated production and perception in use. In this respect, the present account is broadly compatible with ecological and distributed approaches that treat language as an emergent coordinative activity arising through interaction and embodied system dynamics (e.g. Steffensen, 2015). It is proposed here as a process operating within language use rather than as a discrete module or representational component. It does not consist in conscious comparison among alternatives, nor in post-hoc correction, nor in the application of a static hierarchy of abstract constraints. Rather, differentiation arises as phonetic configurations are produced and perceived within a speaker's currently internalised phonological organisation. Although the neuro-motor-perceptual capacities underlying this process are shared across humans, the structural organisation relative to which evaluation occurs is language-specific and historically entrenched. ICE is therefore universal in mechanism but system-relative in its outcomes.

5.2 Operation during production

ICE can be situated within the planning stage of language use during production. Speech production models describe utterances as assembled through successive stages, including lexical selection, phonological encoding, and articulatory preparation (Levelt, 1989). ICE is proposed to operate primarily during phonological encoding and articulatory preparation, where competing phonological candidates are differentiated through the interaction of production, perception, and the speaker's internalised phonological organisation.

When several candidates are available, speakers differentiate them according to how efficiently their configurations integrate into the ongoing production and perception. This process does not involve explicit comparison among alternatives. The outcome is a small difference in integration efficiency within the system. Here, integration refers to the extent to which a candidate form can be coordinated with the speaker's existing phonological organisation across articulatory planning, perceptual recoverability, and system-level pattern compatibility. Forms that are more reproducible within the linguistic system acquire a slight processing advantage and are therefore more likely to recur consistently in subsequent use. This evaluation reflects interacting processing pressures inherent to language use, including alignment with phonological organisation, efficiency of articulatory coordination, and reliability of perceptual identification. These pressures do not operate as a fixed hierarchy but interact dynamically within real-time production and perception.

5.3 ICE in circulation and causal chain of stabilisation

The empirical cases discussed earlier reveal a gap between constructibility and distribution. Morphophonological systems often permit several possible outputs, yet speakers converge consistently on only a subset in usage. This gap indicates that grammatical possibility alone does not determine linguistic outcomes. During language use, ICE differentiates among competing forms relative to the speaker's phonological organisation, introducing asymmetries

in their reproducibility. These asymmetries accumulate through repeated enactment, allowing certain forms to stabilise while others recede.

ICE does not replace existing explanations of stabilisation but complements them by specifying an earlier stage in the causal sequence through which stable patterns emerge. Once asymmetry is established during production, usage-based and exemplar mechanisms amplify these differences through frequency and memory consolidation. Stabilisation therefore reflects the cumulative amplification of differences originating during language use. Accordingly, the account ICE proposes two analytically distinct layers of stabilisation: evaluative stabilisation and distributional stabilisation. Evaluative stabilisation refers to the emergence of system-compatible reproducibility during production, prior to frequency and exemplar accumulation. Distributional stabilisation refers to the subsequent reinforcement of these forms through repeated circulation, frequency, and entrenchment.

ICE therefore locates one source of differentiation in processes operating during production, linking grammatical possibility to distributional outcomes. From this perspective, ICE remains compatible with contemporary usage-based approaches that emphasise propagation, conventionalization, and entrenchment, while proposing that such processes operate upon forms that have already undergone earlier stages of system-level differentiation during language use. The implication of this relationship with existing linguistic frameworks are considered in the following discussion.

6. Discussion

The perspective developed here advances a unified account of the conditions under which grammatically equivalent forms stabilise prior to large-scale frequency differentiation or representational entrenchment, a problem that has remained only partially addressed within phonological models. Rather than treating stabilisation as a consequence of social choice, representational entrenchment, or morphological structure, I propose Internal Coordinative Evaluation (ICE) as a production-based process that differentiates among grammatical possibilities in language use. As defined in 5.1, ICE's coordinative character refers to the dynamic integration of multiple production-level pressures, not to conscious comparison among alternatives or to gestural coordination in the narrow articulatory sense. This shift in perspective does not eliminate social meaning or speaker agency, but instead repositions them as interpretive layers that follow structural differentiation.

The term "evaluation" has appeared in linguistics in multiple senses. In generative traditions, grammars were compared through evaluation metrics (Chomsky, 1965). In Optimality Theory, candidate outputs are evaluated against ranked constraints (Prince and Smolensky, 2004). In sociolinguistics, evaluation refers to speakers' overt commentary on forms (Labov, 1972). ICE differs from all of these. It does not refer to formal comparison among grammars, to constraint-ranking over abstract candidates, or to conscious metalinguistic judgment. Instead, it designates a process that differentiates among competing forms prior to reflection and independent of explicit commentary.

6.1. ICE in relation to phonological and sociolinguistic approaches

Traditional phonology has long acknowledged factors such as ease of articulation and constraint satisfaction (Hayes 2009; Prince and Smolensky, 2004). However, these are typically treated as rule systems or ranked constraint architectures that define well-formed outputs independently of the context of use. Browman and Goldstein (1992) likewise demonstrated that speech production is organised through coordinated gestural dynamics rather than purely symbolic rules. While such approaches illuminate how speech gestures are organised and timed, they do not directly explain why certain grammatical alternatives stabilise while others remain marginal in use. Less explored is why structurally equivalent outputs exhibit divergent stabilisation across contexts. The register data presented in Section 3 bear directly on this gap since *ceṭṭul* satisfies the structural requirements of poetic meter but fails in conversational processing, while *pāṭhaśālaḷu* persists in formal registers through institutional scaffolding but fails in spontaneous speech. These contrasts cannot be derived from context-independent constraint systems alone and thus point to a mechanism sensitive to the conditions under which forms are enacted.

Phonetic theories have emphasized that articulatory and perceptual pressures operate continuously in language use during production (Lindblom, 1990; Ohala, 1993). These accounts successfully explain gradient reduction, assimilation, and variation in realisation, but primarily at the level of segments or features. They do not directly account for why competing well-formed alternatives diverge in stabilisation outcomes. ICE does not deny articulatory pressure, but situates it within system-relative organisation during enactment. The relevant question is therefore not simply which form is easier in isolation, but which form aligns with the active organisation of the system.

Similarly, sociolinguistic models have offered powerful explanations for how linguistic forms acquire social meaning, index identity, and become associated with prestige or stigma (Agha, 2007; Silverstein, 2003). These accounts clarify why speakers attach value to linguistic forms. However, they do not fully explain convergence without explicit ideological alignment or the persistence of low-status forms in everyday production. The cross-demographic uniformity observed earlier suggests a mechanism that differentiates forms prior to social interpretation.

6.2. ICE and usage-based and exemplar accounts

Usage-based accounts demonstrate that repeated exposure strengthens representations through repeated use, making forms easier to access and reproduce (Bybee, 2001, 2010). In this view, stabilisation follows a self-reinforcing cycle in which use increases frequency, frequency reflects accumulated reproducibility, and strengthened representations increase the likelihood of further reproduction. This explains how forms stabilise once frequency differences have emerged. However, it does not account for why grammatically transparent plural forms such as *ceṭṭulu* and *pandulu* are rejected categorically across speakers including

illiterate adults and young children. A purely frequency-based account has nothing to say about the rejection of a form that has never become distributionally established.

Exemplar theory locates differentiation in the accumulation of stored phonetic tokens in memory (Pierrehumbert, 2001). Here, production probabilities emerge from the density of stored exemplars, such that forms heard more frequently exert a stronger probabilistic influence on subsequent production. This theory explains how patterns stabilise through accumulation. However, because it begins with input tokens and their accumulation, it does not explicitly address how certain grammatically possible forms fail to become available as stable inputs in the first place. The selection of which forms become available for storage therefore remains outside its explanatory scope.

Both frameworks acknowledge that early asymmetries may arise prior to large-scale frequency divergence, often attributed to distributed factors such as phonetic ease, analogy, pragmatic fit, semantic nuance, and social alignment. The question is therefore not whether early differentiation occurs, but where in the production process these influences become resolved into reproducible linguistic outcomes.

ICE addresses this question by locating that resolution during language use itself. Rather than treating these influences as independent biases whose cumulative effects eventually favour one form, ICE proposes that they become integrated through a single coordinative evaluative process operating relative to the speaker's existing phonological organisation. The outcome is a subtle asymmetry in reproducibility before frequency-based amplification begins.

A particularly revealing illustration is the *pannu* split in subsection 3.1.2. The surface form *pannu* can mean 'tooth' or 'tax', yet produces different plurals such as *paḷḷu* for 'tooth' and *pannulu* for 'tax'. Usage-based accounts and exemplar theory can describe the endpoint of this divergence by noting that the two forms develop different frequency distributions and exemplar densities across contexts. ICE instead focuses on locating the source of divergence in the evaluative conditions active during production. The 'tooth' sense occurs primarily in conversational contexts where phonological coherence and articulatory coordination favour the compressed form *paḷḷu*, while the 'tax' sense appears only in formal and institutional contexts where the expanded form *pannulu* integrates more readily. The differing production conditions thus provide distinct evaluative environments from which later frequency differences emerge, extending rather than replacing usage-based and exemplar accounts by identifying an earlier stage of coordinative differentiation that precedes both processes.

6.3. Constructibility and Reproducibility in system context

An additional elicitation case provides insight into the relationship between generation and stabilisation. When asked to produce the plural of *paṇḍu*, young speakers produced *paṇḍulu* after a brief hesitation, while adult speakers immediately corrected the form to *paṇḍlu*. The brief hesitation preceding the response indicates competition among alternative candidates in the absence of an entrenched form, rather than the engagement of explicit awareness or metalinguistic reasoning. This pattern points to a distinction between constructibility and

reproducibility. Forms such as *paṇḍulu* are constructible but fail to remain reproducibly integrated once production becomes organised around stabilised patterns.

From the perspective of ICE, this contrast reflects the operation of evaluation under different configurations of the linguistic system rather than its presence or absence. In the absence of entrenched patterns, evaluation may favour structurally transparent constructions. As the system comes to be organised around recurrent patterns, the same evaluative process increasingly favours constructions that align with those patterns, even when they are less transparent. ICE is therefore not a fixed selector of forms, but a dynamic evaluative process whose outputs depend on the current organisation of the system, operating across both early and stabilised stages of language use during production, where it sustains dominant forms and modulates their reproducibility under changing conditions.

One might attribute the contrast between *paṇḍulu* and *paṇḍlu* to articulatory ease or frequency. I argue that neither factor alone is sufficient to account for the observed pattern. If articulatory ease were decisive, forms such as *paṇḍulu* and *ceṭṭulu* should remain viable. But their categorical absence suggests that ease alone does not determine outcomes. Similarly, a frequency-based account explains how *paṇḍlu* persists once established, but does not explain why *paṇḍulu*, a fully constructible form, fails to enter circulation in the first place. This indicates that selection reflects system-level differentiation during production rather than motor efficiency or exposure alone. This process provides a basis for understanding how grammatical possibility becomes distributionally constrained in use, without assuming that all constructible forms are equally available for stabilisation.

6.4. Usage-based refinements and morphological competition: Complementary gaps

The notion of entrenchment itself has long occupied a central place within cognitive and usage-based linguistics (e.g. Langacker, 1987). Recent developments in usage-based linguistics have significantly refined the role of frequency in shaping linguistic structure. Rather than treating frequency as a simple determinant of entrenchment, contemporary approaches increasingly relate entrenchment to broader cognitive and interactional processes (Schmid, 2015). Likewise, Divjak (2019) argues that frequency effects are mediated by factors such as attention, memory, and categorisation, such that linguistic representations emerge from the interaction between usage patterns and domain-general learning mechanisms. Similarly, Diessel (2019) conceptualises grammar as a dynamically evolving network of interconnected constructions, in which structure arises through repeated usage and the strengthening of associative links between forms.

Related construction-based approaches likewise model linguistic structure as emerging from recurrent form-meaning pairings and their distribution across usage events (e.g. Goldberg, 2006, 2019). Recent work in diachronic construction grammar has similarly emphasized how recurrent usage patterns shape the emergence and stabilisation of constructions over time (Hilpert, 2013, 2021). Alongside these developments, broader frameworks such as Schmid (2020) have further extended usage-based theory by embedding frequency within processes of interaction, conventionalization, and sedimentation. In these accounts, linguistic structure is

not only shaped by repeated usage but also by the alignment of speakers within a social system, with stabilisation emerging from patterns of propagation and shared conventions over time.

In parallel, research on morphological competition and variation has demonstrated that linguistic systems are often characterized by uncertainty, coexistence, and absence rather than uniform stability. Work by Bermel and Nikolaev (2022), for example, shows that speakers may exhibit probabilistic behaviour or avoid particular forms altogether, indicating that not all paradigmatic slots are fully resolved. Similarly, studies of overabundance by Thornton (2011) demonstrate that multiple forms may coexist within a system, each serving the same grammatical function without a single form necessarily dominating. Conversely, research on defectivity, such as O'Neill (2025), highlights that some forms fail to emerge or remain systematically avoided, revealing structured gaps within paradigms. Related notions of propagation (MacWhinney, 2022), diffusion (Milroy, 1992), conventionalisation (Traugott and Trousdale, 2013; Traugott and Hilpert, 2020) further describe how forms spread through social networks and stabilise as shared conventions, while leaving open the question of how particular forms become reproducibly integrated prior to such circulation.

Taken together, these approaches provide increasingly detailed accounts of how linguistic forms are distributed, propagated, and organised across speakers and systems. They explain how representations are strengthened through experience, how networks of constructions emerge, and how variation and uncertainty are maintained within linguistic systems. However, despite these advances, such models primarily operate on representations that are already available for use, accumulation, or competition. They focus on how forms are strengthened, propagated, or distributed, rather than on how specific forms are internally resolved at the initial stages of use. In particular, they do not explicitly model the transformation of incoming forms into system-compatible representations prior to their participation in processes of entrenchment, network formation, or competition. This gap becomes especially evident in cases where forms exhibit stable retention even under minimal exposure, or where multiple forms coexist without converging on a single outcome, or where certain forms fail to stabilise altogether. This pre-entrenchment gap manifests empirically in the *pannu* ('tooth'/'tax') split discussed in 6.2, where an identical lexical input yields two register-restricted plurals prior to any frequency divergence between them. Usage-based models can describe such distributions once established, but do not specify an explicit mechanism through which domain- or register-specific differentiation emerges during enactment. More importantly they do not explicitly model why *pa!lu* stabilises in everyday speech while *pannulu* remains restricted to formal registers.

ICE addresses this issue by introducing a stage of system-level evaluation that operates prior to frequency-based effects. Rather than treating frequency alone as the primary driver of stabilisation, ICE proposes that incoming forms are first evaluated and reshaped into system-compatible representations. Only those forms that achieve such compatibility become available for repeated use, propagation, and entrenchment. In this sense, ICE complements existing usage-based and competition-based accounts by specifying the internal resolution process that precedes and underlies the distributional dynamics they describe.

6.5. ICE and minimal exposure: Stability without frequency

The explanatory limits of frequency, usage-, and variation-based models become particularly visible in conditions of minimal exposure. Here, ICE reframes the relationship between stabilisation and frequency. It argues that stability is not maintained through continuous reinforcement alone, but through structural fit within the system. Most usage-based models discussed here treat entrenchment as emerging through repeated usage and reinforcement, with frequency strengthening representations over time (Bybee, 2010; Divjak 2019). Under such accounts, rarely encountered forms might be expected to exhibit weaker representations, potentially leading to instability, variation, or even forgetting. However, speakers can often retain and reproduce forms accurately even after minimal exposure and extended gaps in use. Such cases are not easily accounted for if frequency is treated as the primary determinant of stability. A similar issue arises for exemplar-based models. When only one or a few exemplars are available, no dense memory cloud can be formed, and the resulting representation would be expected to be relatively fragile, potentially leading to noise or distortion in recall. Yet speakers frequently reproduce such forms with striking consistency even under these conditions.

ICE offers a different perspective. It proposes that what is stored is not the raw input, but a system-evaluated and internally compatible representation. In terms introduced in 5.3, this corresponds to evaluative stabilisation, in which a form achieves reproducible coordinative fit within the linguistic system prior to large-scale distributional reinforcement. Stability, in this view, does not depend on repeated reinforcement, but on the degree to which a form is structurally resolved within the system. Once successfully evaluated and integrated, a form can remain stable even after minimal exposure. For example, an English form such as *apple* may be encountered only briefly by a Telugu speaker, yet subsequently reproduced decades later as *yāpil*, not because the original acoustic input was repeatedly reinforced, but because the form had already undergone system-relative evaluative restructuring during initial exposure. What persists in memory is therefore not the raw input token, but the internally resolved representation that achieved coordinative compatibility within the speaker's phonological organisation. In other words, the spread of a form is not driven by frequency alone in the sense of repeated variation and competition among alternatives, but by the transmission of a system-compatible representation that has already been stabilised through evaluation. This restructuring illustrates evaluative stabilisation prior to later distributional stabilisation through repeated use and circulation. Although such restructuring becomes especially visible in contact-induced adaptation, the same evaluative process is proposed to operate within native linguistic systems whenever multiple constructible alternatives compete for reproducible integration. This makes ICE a general mechanism applicable to both external adaptation and internal variation. From this perspective, frequency may be better understood as reflecting the distributional amplification of system-compatible forms across speakers and communities, rather than as the primary mechanism responsible for their stability within individual cognitive systems.

7. Theoretical Implications

Several implications follow from the ICE process proposed in this study.

First, the ICE account reframes debates about which regional form is better or more correct and clarifies why speakers perceive certain linguistic forms as more natural or aesthetically satisfying than their alternatives. Within this perspective, the judgments that arise during ordinary speech reflect how configurations integrate with the phonological organisation active in language use during production. Alternative forms that align with this organisation are reproduced with greater coordinative ease and therefore more likely to be experienced as natural, whereas competing alternatives that impose greater coordination demands are perceived as marked. Such judgments of correctness or elegance can thus be understood as reflections of this evaluative process, rather than being attributable solely to social or ideological factors.

Second, ICE provides a process-level account of linguistic stabilisation. Grammar specifies the range of constructible possibilities available within a language, but ICE proposes a mechanism through which those possibilities become differentially reproducible during use. Once these early asymmetries emerge, repeated reproduction allows established usage-based mechanisms, including frequency, propagation, and entrenchment, to reinforce them over time. Stabilisation therefore reflects both early coordinative differentiation and subsequent distributional amplification, rather than either process alone.

Third, the present account suggests that frequency should be interpreted mainly as an outcome of earlier production-level differentiation rather than as its sole cause. This perspective helps explain why constructible forms sometimes fail to circulate despite being grammatically available, why contact varieties may stabilise forms that differ from those predicted by simple exposure, and why register-specific forms persist under different production conditions. Rather than replacing frequency-based explanations, ICE identifies an earlier stage of differentiation that contributes to the asymmetries upon which such explanations subsequently operate. This perspective therefore complements existing accounts by extending the causal sequence through which linguistic forms become stabilised.

More broadly, ICE shifts attention from language as a static inventory of forms towards language as an organised process of production. On this view, linguistic variation and change emerge not only through social transmission and repeated exposure, but also through recurrent differences in how competing forms are evaluated during language use.

8. Conclusion and future directions

The present account does not claim to directly observe internal coordinative evaluation as a discrete process, but instead infers its operation from recurrent patterns of differential production observed across multiple linguistic domains. If Internal Coordinative Evaluation plays the role proposed here, several consequences follow. First, forms that integrate more smoothly into ongoing patterns of language use should tend to become more reproducible during production and subsequently circulate more widely through repeated use, even when

multiple grammatically possible alternatives exist. Second, artificially introduced expanded forms should show lower rates of spontaneous adoption unless they align with existing phonological organisation. Third, in contact situations, borrowed forms should converge predictably toward the syllable structure and timing patterns of the recipient language rather than remain in structurally unsettled states. Finally, high-frequency embodied domains may exhibit comparatively stronger compression tendencies than low-frequency abstract domains, reflecting cumulative reinforcement through repeated use. These expectations are testable through corpus analyses tracing the earliest emergence of competing forms, controlled production experiments with competing forms, and reaction-time studies examining reproducibility asymmetries during production. If ICE operates as proposed, competing forms should exhibit measurable differences in initiation latency or error rates under controlled tasks.

This study has argued that grammatical possibility alone does not determine which forms become established in language use. The empirical cases presented here suggest that competing grammatical possibilities become differentially reproducible before large-scale frequency divergence, entrenchment, or conventionalisation emerge. ICE is proposed as the production-level process linking grammatical possibility with subsequent distributional outcomes.

Although the empirical illustrations are drawn from Telugu, the argument is intended as a general hypothesis about language rather than a language-specific account. The cross-linguistic examples discussed here suggest that similar production-level differentiation may operate whenever competing forms are integrated into a speaker's phonological organisation, regardless of the particular language involved. The present account remains intentionally modest in scope. It does not specify the precise computational or neurocognitive mechanisms underlying internal coordinative evaluation, nor does it claim that the proposed process alone explains all aspects of linguistic variation and change. Rather, it identifies an explanatory stage that received comparatively little explicit theoretical attention within existing theories and formulates it as testable hypothesis. Future work can refine its underlying mechanisms, identify measurable correlates of internal coordinative evaluation, and assess its cross-linguistic generality.

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